

Hamza Faisal

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FULL STACK ENGINEER | REAL-TIME SYSTEMS & IoT ANALYTICS

Full Stack Engineer with over 4 years of professional experience building real-time, data-intensive web applications across IoT, fintech, and precision agriculture domains. Specializes in TypeScript, JavaScript, Node.js, React/Next.js, SQL and NoSQL databases, and AWS-based infrastructure. Experienced developing REST APIs, scalable microservices, responsive user interfaces, CI/CD workflows, and event-driven systems with a consistent track record of delivering production-grade systems, measurable performance improvements, and scalable microservice architectures. Published researcher in AI-driven edge computing (Springer, 2025) with a B.S. in Computer Science from Valparaiso University.

WORK EXPERIENCE

Chicago Dryer Company

March 2025 - Present

Full Stack Engineer | *Typescript, Node.js, AWS & Full-Stack Systems*

Chicago, IL, USA (On-site)

- Developed a scalable industrial IoT analytics platform using **Node.js**, **MongoDB**, and **Next.js** to track real-time machine states, operator performance, and production metrics, reducing diagnostics time by **70%** and supporting **1M+** datapoints daily.
- Refactored monolithic backend into modular microservices with machine-specific data ingestion and simulation interfaces using **Express.js** and **DataFusion** protocols, boosting processing speed by **4x** and enabling independent testing.
- Deployed and maintained **Dockerized Node.js** services in **AWS-hosted Linux** environments, configuring **Nginx** reverse proxying, environment-specific application settings, logging, and **CI/CD-driven** releases for reliable production deployments.
- Optimized complex **MongoDB** pipelines and backend APIs for analytics routes, eliminating **N+1** queries and reducing response times from **9s** to **<1.5s** through aggregation refactors and indexing strategies.

MarketWise

Nov 2023 - Jan 2025

Frontend Engineer

Dubai, UAE (Remote)

- Built and maintained **Angular** applications in **TypeScript**, developing reusable components, services, reactive forms, and REST API integrations for responsive customer-facing workflows.
- Engineered a low-latency, real-time UI using **WebSockets**, **Next.js**, and **Apollo Client**, achieving **sub-100ms** response times. Integrated **GraphQL** APIs via **Express.js** for efficient state synchronization.
- Built a secure signaling layer using **Socket.io** to establish **WebRTC** connections and handle session negotiation.
- Integrated **WebRTC** for peer-to-peer real-time video and audio communication, enabling direct browser-based calling with minimal latency.
- Automated application build, test, and deployment workflows using **GitHub Actions**, integrating validation into pull requests and production releases.

Adson As

Dec 2022 - July 2023

Frontend Developer

London, UK (Remote)

- Designed a modular UI component system in **React.js** and **Next.js**, leveraging **TypeScript** and **Tailwind CSS** to enhance performance and maintainability across web and mobile apps.
- Collaborated with the Design Lead to craft animation-rich UI components using **Framer Motion**, **Tailwind CSS**, and **React.js**. Designed custom sliders, selectors, and dynamic elements to enhance user experience.
- Designed and implemented secure payment UI workflows integrated with **REST APIs** and **Webhooks**, enhancing transaction reliability and error handling.
- Developed **Kafka** consumers and producers in **Node.js** to support asynchronous messaging pipelines for critical telemetry and state data.

Lyftyfy

March 2021 – July 2022

Frontend Developer

Hamburg, Germany (Remote)

- Refactored legacy **jQuery** and **AngularJS** components into a modern **React** architecture, utilizing **React Hooks** for stateful logic, **Redux** for global state management, and **Webpack** for optimized bundling and performance.
- Built responsive dashboards with interactive charts and data visualizations using **Victory.js** and **D3.js**, enabling real-time analytics for end-users.
- Optimized **Lighthouse** scores by **30%** using **Webpack** tree shaking, **Babel** transpilation, and asset minification.
- Collaborated with backend lead to enhance database performance in **Node.js (Express.js)**, analyzing and optimizing raw **SQL** queries in **PostgreSQL** with strategic indexing and refactoring, achieving over 2 times speed improvement for critical queries.

PROJECT EXPERIENCE

Valparaiso University

Aug 2024 – Dec 2024

Frontend Engineer and Researcher

Valparaiso, IN

- Crafted and developed a cross-platform mobile application using **React Native** and **Expo**, enabling real-time crop monitoring and management, improving crop yield predictions by 30% through AI-driven insights.

- Developed complex interactive UI components, including **geospatial visualizations with Mapbox and Leaflet**, improving user decision-making.
- Implemented **MongoDB** for real-time database management and authentication, improving system reliability by **20%**. Developed **GraphQL** API endpoints with **Node.js (Express.js)** for efficient data synchronization and scalable microservices.
- Designed and built interactive UI dashboards using **Victory Native** and **Skia**, enabling real-time data visualization and predictive analytics for smart agriculture.

EDUCATION

Valparaiso University

Bachelor of Science in Computer Science

Valparaiso, IN

Graduation Date: Dec 2024

- College of Arts and Sciences Dean's List Award
- International Presidential Scholarship Recipient
- **EnviroSense: AI-Driven Microclimate Control for Sustainable Agriculture Using Edge Computing** – Published in Lecture Notes in Networks and Systems (Springer) at SAI Computing Conference 2025, London, UK.

TECHNICAL SKILLS

Programming Languages: JavaScript, TypeScript, Python, Java, Ruby on Rails

Tools & Frameworks: React.js, React Native, Redux, Next.js, Angular, HTML, CSS, SCSS, Tailwind CSS, Material UI, REST, AWS, Kafka, Docker, Linux (Ubuntu/Debian), WebSocket, Socket.io, APIs, GraphQL, Apollo, Git, Firebase, MySQL, PostgreSQL, Jenkins, Webpack, Babel, Nginx, Node.js, Express.js, Jest, Cypress, Mocha

Other: CI/CD pipelines, Agile/Scrum methodologies, UI/UX design principles, Figma/Adobe XD